

Computation and Boundary Evidence for Scientific Software Repair

Rajarshi Ghoshal

rajarshi.ghoshal1@gmail.com

PhD applicant research task — Graz University of Technology

Abstract

A repair agent with shell tools must reconstruct a repository’s computational structure from text search. CBE (computation and boundary evidence) prepares that structure once and lets the agent query it during repair, under the same model and budget. On 89 held-out SWE-bench Science tasks ($k=3$), both arms solve 60 attempts (22.9% vs. 23.1%). Aggregate parity masks a discipline-level split: mechanics +22.2 pp, materials +9.7 pp, astronomy +12.4 pp; biomedical engineering −16.7 pp, biology −9.5 pp. CBE generates 13.7% fewer output tokens. About a fifth of tasks flip between identical runs, bounding what any single-run comparison can resolve.

1 Introduction

Repairing a composite-laminate stiffness calculation requires changing both the ply iteration and the accumulation formula. Expanding one without the other passes the public check but fails the private tests. The patch is short; finding it means knowing which expression produces the returned matrix and what calls cross the repository boundary.

We give the agent that knowledge as a prepared store and test whether it changes outcomes. CBE exposes source-linked computation structure; the baseline reconstructs the same information from `grep` and file reads. Same model, same budget.

Contributions: the preparation pipeline, the query interface, and a replicated evaluation that separates discipline-level differences from aggregate noise.

CBE draws on traceability (Niu et al., 2023b,a, 2025; Pan et al., 2025), code property graphs (Yamaguchi et al., 2014), scientific model extraction (Pyarelal et al., 2020; ML4AI Team, 2024), and incremental parsing (Brunsfeld et al., 2024). We evaluate on SWE-bench Science (Xu et al., 2026) with DeepSeek V4.1 Flash (DeepSeek, 2026); SWE-bench (Jimenez et al., 2024; Yang et al., 2024) and SciCode (Tian et al., 2024) are related benchmarks.

2 What CBE provides

Prepare. Run the public `reproduce.py` under an observer. Observed functions guide file selection. Static extraction adds expressions, conditions, definitions, and call boundaries. No LLM call. Median duration: 73 s of the 1800 s budget.

Table 1. Evidence fields, from task 103’s saved records.

Record	Content
Computation	Scope, expressions, result anchors: accumulation at line 1003 → return ABD at 1015.
Condition	<code>len(z0) == len(z1)</code> at line 987.
Definition	Documented public behavior (not execution trace).
Boundary	<code>np.radians(thetad)</code> at line 979: provider <code>numpy</code> , caller-side scope.

Query interface (alongside ordinary shell)

<code>find</code>	Targets and source excerpts.
<code>inspect</code>	Relationships, source, definitions, or contracts.
<code>note</code>	Optional working model; never gates a repair.

Represent. Tree-sitter (Brunsfeld et al., 2024) parses Python ASTs to extract function scopes, assignment targets, binary expressions, return anchors, and conditional branches. Joern handles C, C++, Fortran, and Cython where available; explicit fallback otherwise. External calls are labelled with their provider as interface assumptions, not correctness guarantees. Unresolved callees stay `unknown_external`.

Query and repair. The agent accesses three endpoints alongside shell utilities: `find` searches extracted symbols, expressions, and docstrings; `inspect` retrieves computation scopes, dependencies, and boundary assumptions; `note` records an explicit working model into a durable scratchpad without gating repair actions. Inspection dynamically refreshes edited source lines, while preparation traces remain fixed.

3 Protocol

119 tasks, 98 repositories, 20 disciplines (Xu et al., 2026). 30 tasks held for development, 89 locked for evaluation.

Table 2. Shared conditions.

Component	Detail
Model	DeepSeek V4.1 Flash (DeepSeek, 2026); temperature 0; PYTHONHASHSEED=0.
Replicates	89 tasks $\times$ 3 runs = 267 scheduled per arm; 522 verified.
Budget	1800 s, including preparation.
Verification	Separate container, 1800 s, runs after repair ends. Private tests not visible.
Network	None.
Baseline	Shell tools + public task material.
CBE	Same + evidence store + three query endpoints.

A solve is the official verifier reward. Missing verdicts excluded, not failures. Paired test: bootstrap per-task rate differences (20k resamples, fixed seed). Sign test excludes ties.

Each trial executes in an isolated Docker container with 8 CPUs and 16 GiB RAM under a strict 1800 s total budget. Evaluation applies the final git patch in a clean container and executes the unobserved private test suite.

4 Parity, fewer tokens

Table 3. Both arms solve 60. CBE generates 13.7% fewer output tokens.

Metric	Baseline	CBE
Scheduled attempts	267	267
Exact repairs / verdicts	60/262	60/260
Exact-repair rate	22.9%	23.1%
Private tests passed	2315/3167	2226/3148
Output tokens (M)	12.01	10.36
Input tokens (M)	838.0	878.3
Mean work per attempt (s)	1236.6	1306.9

Paired task-rate diff: +0.75 pp; 95% bootstrap $[-3.4, +4.9]$ pp; sign $p = 1.00$.

74 tasks tie, 7 favour CBE, 8 favour baseline. Output drops 13.7% (10.36M vs. 12.01M tokens). Input rises 4.8%. Work takes 70 s longer. The agent searches less — it either finds the right computation or anchors on the wrong one.

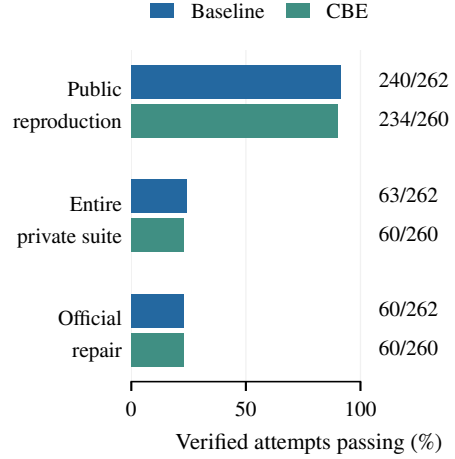


Figure 1. Most attempts pass the public check without satisfying the full verifier.

CBE avoids speculative file reading and iterative grep pipelines. By querying pre-indexed computation graphs and boundary signatures directly, the agent issues fewer exploratory turns, reducing total output generation across verified attempts.

Hidden partial changes. Of 74 exact ties, 30 differ in private-test coverage (Table 4).

Table 4. Exact-rate ties crossed with private-test coverage.

Exact rate	Private-test coverage		
	Baseline higher	Equal	CBE higher
Baseline higher	7	0	1
Equal	21	44	9
CBE higher	0	0	7

Of the 74 exact ties, 30 show divergence on held-out private tests: CBE improves test coverage in 9 tasks without changing binary solve status, while baseline achieves higher coverage on 21. For tasks that both arms solve, coverage is identical in 44 instances.

5 Discipline split

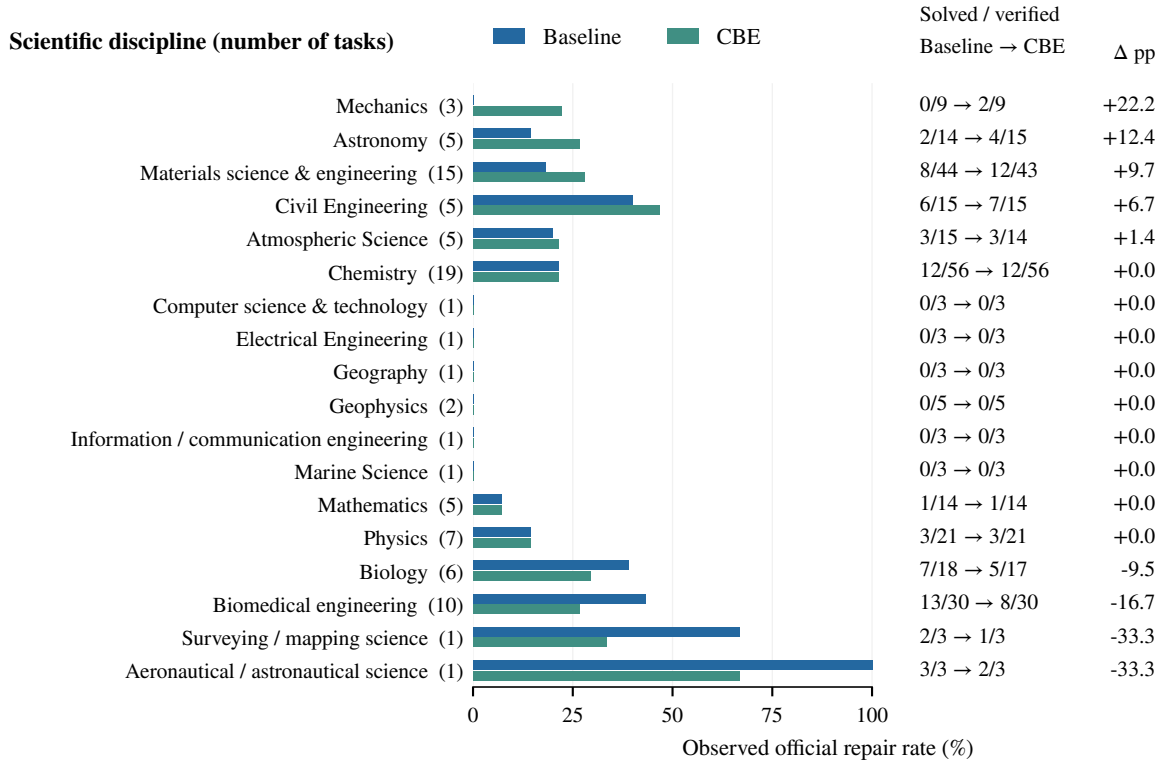


Figure 2. Parity hides opposing domain effects. All 18 represented disciplines; task counts in parentheses; right column is the percentage-point change.

Mechanics: +22.2 pp, entirely task 103 (pyNas-tran). The agent inspects the ply-angle computation and its numpy boundaries, then edits — 2/3 solves vs. 0 baseline. Materials: +9.7 pp across three tasks. Astronomy: +12.4 pp.

Biomedical engineering: −16.7 pp across four tasks. Task 022 (nilearn): baseline solves 2/3, CBE 0/3. The trajectory reads projection mathematics before touching the atlas-label mapping. Biology: −9.5 pp. The two −33.3 pp domains are single-tons.

Chemistry (19 tasks), physics (7) and mathematics (5) do not move. No interaction with the ablation flag: −0.3 pp on ablated tasks, +1.4 pp on knowledge-provided ones. 81 of 89 tasks are pure Python, so language strata are uninformative.

When the defect is a computation, structure finds it. When an ordinary bug sits under scientific prose, structure distracts.

Replication stability. Most tasks never solve or always solve across repeated runs; about a fifth flip within an arm (Figure 3).

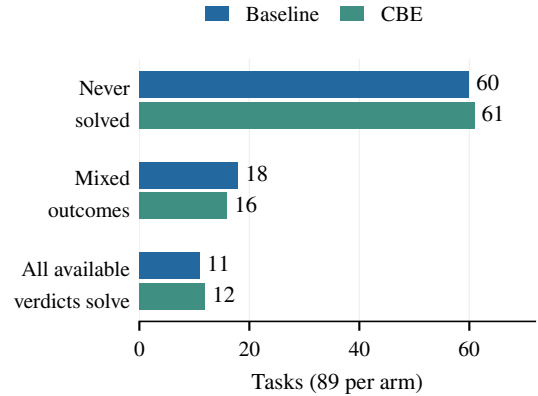


Figure 3. Most tasks never solved or always solved. About a fifth flip within an arm.

Across the 89 tasks, 60 never solve in either arm, 11–12 solve in all three runs, and 16–18 yield mixed outcomes. This within-arm stochasticity bounds single-run benchmark resolution and shows aggregate parity is not an artifact of run-to-run noise.

6 Case studies

Task / repository	Discipline	Baseline			CBE			Solved / verified
		Run 1	Run 2	Run 3	Run 1	Run 2	Run 3	
074 DeePTB	Materials	9/10	9/10	2/10	10/10	10/10	—	0/3 → 2/2
042 sunpy	Astronomy	15/16	15/16	6/16	16/16	7/16	16/16	0/3 → 2/3
103 pyNastran	Mechanics	2/4	2/4	2/4	4/4	4/4	2/4	0/3 → 2/3
067 xclim	Atmospheric	6/6	6/6	3/6	6/6	6/6	—	2/3 → 2/2
020 deepmd-kit	Materials	10/12	10/12	11/12	10/12	12/12	10/12	0/3 → 1/3
084 dpdata	Materials	5/7	5/7	7/7	7/7	5/7	7/7	1/3 → 2/3
116 Pynite	Civil eng.	85/93	85/93	85/93	85/93	93/93	85/93	0/3 → 1/3
029 biopython	Biology	29/29	14/29	27/29	14/29	14/29	14/29	1/3 → 0/3
053 MONAI	Biomedical	2/6	6/6	2/6	5/6	5/6	2/6	1/3 → 0/3
063 tskit	Biology	9/18	18/18	7/18	7/18	9/18	—	1/3 → 0/2
087 dipy	Biomedical	5/5	0/5	1/5	0/5	2/5	2/5	1/3 → 0/3
093 MONAI	Biomedical	4/4	2/4	2/4	2/4	2/4	2/4	1/3 → 0/3
098 satpy	Surveying	11/11	8/11	11/11	8/11	11/11	8/11	2/3 → 1/3
056 poliastro	Aeronautical	10/10	10/10	10/10	10/10	10/10	1/10	3/3 → 2/3
022 nilearn	Biomedical	12/12	12/12	10/12	4/12	11/12	11/12	2/3 → 0/3

Cell: private tests passed / collected. Outlined + bold: official repair. Dash: no verdict.

Figure 4. Per-run private-test fractions for every task whose solve rate differs between arms. Outlined bold cells are official solves; bold task labels are the two cases below.

Table 5. What changed in the patches.

Task	Baseline patches	CBE patches	What it shows
103 pyNastran	Expand ply iteration, keep original B/D accumulation. All three pass the public check.	Two change both material-reference expansion and B/D expressions; the third keeps the old accumulation.	The public check undertests. Covering the full computation, not just the example, separates the two complete patches from the three partial ones.
022 nilearn	Two add a label-aggregation option without altering the existing nearest-neighbour path.	Two add label-aware aggregation but also change how the existing path is selected or computed; 11/12 private tests pass.	Adding a capability while preserving the existing interface is the obligation. CBE’s trajectory reads projection maths instead of the interface contract.

Task 103: covering the full computation, not just the example, is the repair obligation. Task 022: preserving the existing interface while adding the new case. CBE helped with the first and hurt the second.

Data and reproduction. Full trajectories, unified git diffs, agent transcripts, and verifier receipts are archived in the repository. The evaluation environment is fully deterministic under pinned container images and fixed random seeds.

7 Limitations and conclusion

One model, one effort, one budget. Domain analyses are descriptive; singleton domains carry no weight. The store is approximate: unparseable expressions are recorded as unsupported, external calls are labelled rather than verified. No ablations isolate which part of CBE matters.

CBE reaches parity, generates fewer tokens, and splits the benchmark by discipline. Scientific structure helps when the defect is a computation and hurts when it is ordinary code.

Artifacts and AI use. The author designed the study; LLM assistants (Oh My Pi, ChatGPT) helped implement the harness and revise prose. DeepSeek V4.1 Flash (DeepSeek, 2026) was the only repair model.

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